

SW Requirements Analysis & Specifications

Ch-3

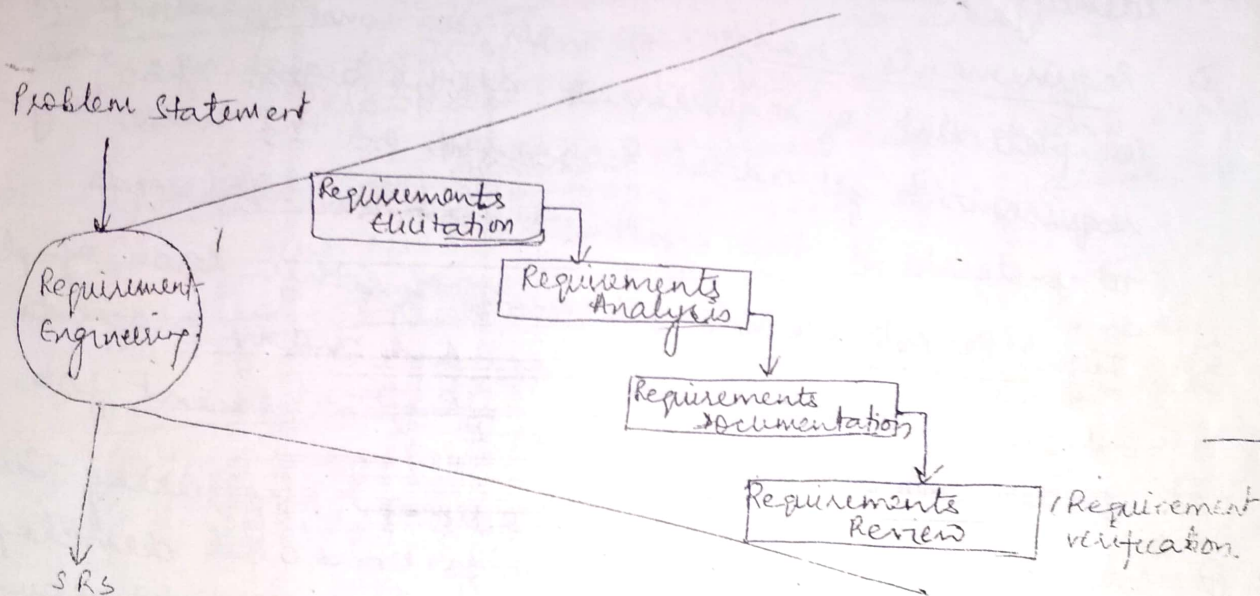
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Requirements Engineering -

Requirements describe "what of a system" & not the "how" of the system.

Crucial steps for Requirements Engg.



Requirements Elicitation - Requirements are gathered as per requirements of the customer & existing system processes, if available.

Requirements Analysis - The requirements are analyzed in order to identify inconsistencies, defects, omissions.

Requirements Documentation - The document is called as SW Requirements Specification (SRS).

Requirements Review - The review process is carried out to improve the quality of the SRS. It's also called as requirements verification).

Difficulties faced in carrying out Requirement Engineering.

- 1) Requirements are difficult to uncover - It is impossible to define all the requirements in the beginning. $\therefore$ users & developers must resort to trial & error to identify problems & solutions.
- 2) Requirements Change - No user can come up with a complete list of requirements at the beginning, the requirements get added & changed as the user begins to understand the system & his other real needs.
- 3) Tight Project Schedule - Because of either lack of plan or unreasonable customer demand, many projects start with insufficient time to do a decent job.
- 4) Communication barriers - Requirement engineering is communication intensive activity. Users and developers have different vocabularies, professional backgrounds, tastes $\therefore$ it may lead to misunderstanding & confusion.
- 5) Market-Driven s/w development - Many of the s/w development is today market driven, developed to satisfy anonymous customers & to keep them coming back to buy upgrades.
- 6) Lack of Resources - There may not be enough resource to build s/w that can do everything the customer wants.

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Requirement

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Type of Requirements

different types of requirements are:-

Known Requirements - something or some requirements that the customer believes to be implemented

Unknown Requirements - Requirements forgotten by the customer becoz they are not needed right now or needed by another customer

3) Undreamt Requirement - Customer may not be able to think of new requirements due to limited domain knowledge.

Functional Requirements - these requirements define what the s/w has to do. They are often called as Product features.

Non-functional Requirements - these are mostly quality requirements that describe how well the s/w does what it has to do. These include specifications of desired performance, availability, reliability, usability and flexibility. ~~these~~ the non-functional requirements for developers are maintainability, portability & testability.

Requirements Elicitation.
Requirements are gathered so as to find out what the user's really need and it is done through an effective customer-developer partnership.

Requirements are gathered by asking questions, writing down the answers, asking other questions, etc. Thus, requirements gathering is the most communication intensive activity of s/w development.

The various techniques used are:-

- 1) Interviews - After defining problem statement, a meeting is arranged with the customer. During the meeting or interview, both the parties understand each other. The main objective is to understand customer's expectation from the s/w.

Selection of Customer & interview

Every customer/user can't be interviewed. ∴ representatives from groups must be selected based on their technical expertise, domain knowledge, credibility & accessibility. Several groups which may be considered are:-

- 1) Entry level personnel - (freshers) - they have sufficient domain knowledge & expertise, fresh ideas & different views.
- 2) Mid-level customers - they have better domain knowledge & experience of the project. They know sensitive, complex & critical areas of the project. (Project managers)
- 3) Managers or other customers - Higher level mgmt officers like Vice-Presidents, General Managers, managing Director should be interviewed. Their expectations must be fulfilled.
- 4) Users of the software - This group is the most imp. becoz they will spend most of the time interacting with the s/w.

1. Types of Questions

(3)

Simple & short questions should be asked.

2) Brainstorming Sessions.

these are type of group discussions which lead to new ideas & help to promote creative thinking & provides platform to share views, apprehensions, expectations & difficulties of implementation.

3) Facilitated Application Specification Technique (FAST)

this is similar to brainstorming sessions & the objective is to bridge the expectation gap - a difference b/w what developers think they are supposed to build & what customers think they are going to get. This approach encourages the creation of a joint team of customers & developers who work together to understand the expectations & propose a set of requirements.

FAST session preparations } study yourself.
" " activities }

4) Quality function deployment

it is a quality mgmt technique that helps to incorporate the voice of the customer. The voice is then translated into technical requirements & documented into an SRS document.

3 types of requirements are identified -

- 1) Normal Requirements - the objectives & goals of the proposed s/w are discussed with the customer. If the requirements are fulfilled then the customer is satisfied.

Use Cases - actors & the goal satisfies the goal + this + (interaction) with the system

2) Expected Requirements - These requirements are implicit to the s/w product & may be so obvious that the customer may not explicitly state them. Eg:- protection from unauthorised access, some warning system for wrong entry.

3) Exciting requirements - Some features go beyond the customer's expectations & prove to be very satisfying when present. Eg:- E-mail generated to the administrator if unauthorised access is noticed, etc.

4) The Use Case Approach.

This approach ~~uses~~ uses a combination of text & pictures in order to improve the understanding of the requirements. They only give the functional view of the system.

Use cases are structured outline or templates for the description of user requirements, modeled in a structured language like English.
The components ^{used} are:

1) Actor - It is an external agent, lies outside the system model, but interacts with it in some way. An actor may be a person, machine or an info system that is external to the system model. Eg:- Customers, Users, External devices or any external entity interacting with the system are actors.

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Use Cases - It describes the sequence of interactions b/w actors & the system necessary to deliver the services that satisfies the goal.

Thus, the use cases capture who (actor) does what (interaction) with the system, for what purpose (goal), without dealing with system internals.

use case template

1) Brief Description

2) Actors

3) Flow of events

1) Basic Flow

2) Alternative Flow

4) Special Requirements

5) Pre-Conditions

6) Post Conditions

7) Extension Points

1) Introduction

2) Actors

3) Pre-Condition

4) Post-Condition

5) Flow of events

1) Basic Flow

2) Alternative Flow

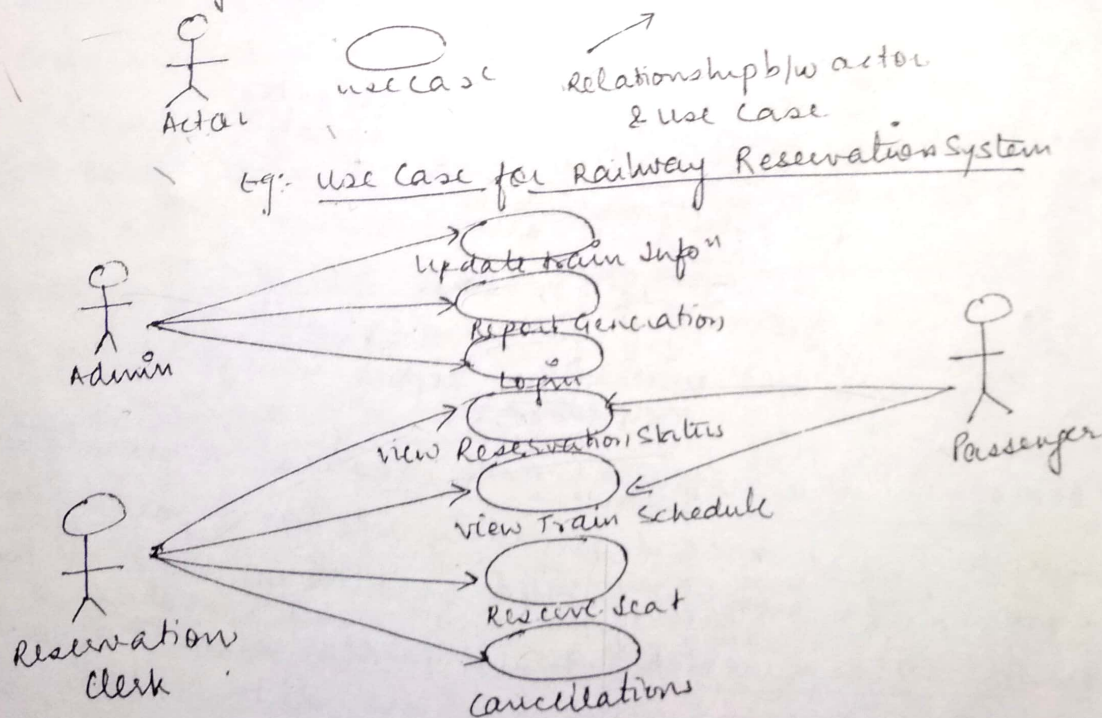
6) Special Requirements

7) Related Use Cases

OR

use case diagrams -

they visually represent what happens when an actor interacts with the system.

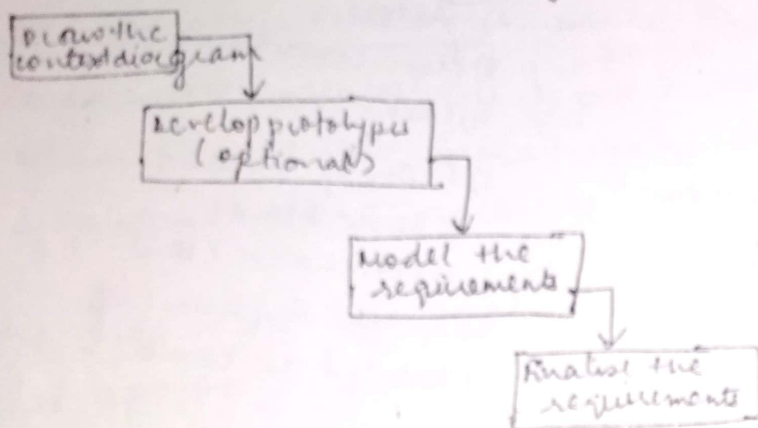


Finalise the documents
Data Flow
Level 0 & 1
Data Dictionary

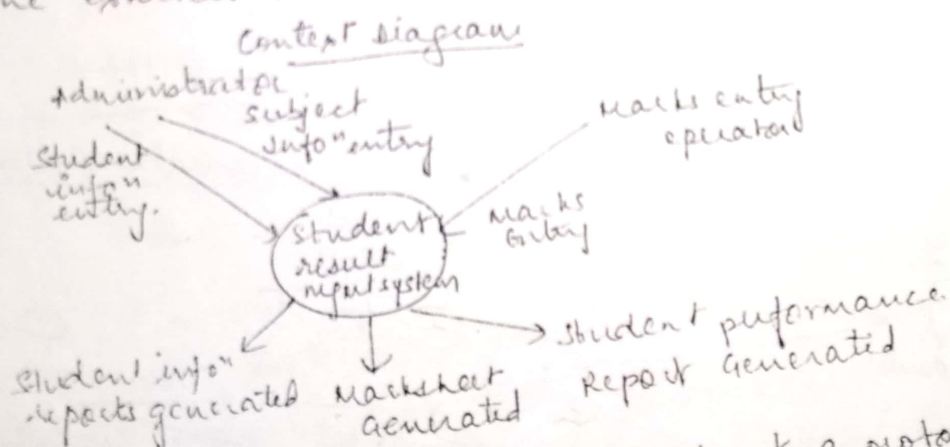
Requirement Analysis

after gathering the requirements, the requirements are analyzed, refined & scrutinised in order to make it consistent & unambiguous.

Requirement Analysis Steps



1) Draw the context diagram - context diagram is a simple model that defines the boundaries & interfaces of the proposed system with the external world.



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- 2) Development of a prototype - To construct a prototype, something that looks & preferably acts like a part of the system. The prototype should be built quickly & at low cost.
- 3) Model the requirements - This process consists of various graphical representation of functions, data entities, external entities & relationship b/w them. Such models include Data Flow Diagrams, ER diagrams, Data Dictionaries, etc.

5) Finalize the requirements - Requirements are finalised & documented into SRS document.

Data Flow Diagram -

Level 0 & Level 1

Data Dictionary

E-R Diagram

Degree of Relationship

Attributes

} Read user.

Requirements Documentation

SRS - is a specification for a particular s/w product, program or set of programs that performs certain functions in a specific environment.

Nature of SRS

the basic issues that should be addressed in SRS are:-

- 1) Functionality - what s/w is supposed to do
- 2) External Interfaces -
- 3) Performance - what is the speed, availability, response & recovery time, etc.
- 4) ~~as per~~ Attributes - what are ~~proposed~~ considerations for portability, correctness, maintainability, ~~etc~~ security, etc.
- 5) Design constraints imposed on an Implementation - policies of database Integrity, Resource limits, etc.

Characteristics of a Good SRS.

- 1) Correct - every requirement stated in it should be met in the s/w.
- 2) Unambiguous - every requirement shall have one/unique interpretation.

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- 1) Correct - Every requirement stated in it should be met in the s/w.
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Complete -

- 4) Consistent - NO subset of individual requirements described in it has conflict.
- 5) Ranked for importance/stability - each requirement described should have importance/stability while operating in a particular environment. (Setting priorities).
- 6) Verifiable - ~~person~~ person/machine can check whether the S/W meets the requirements.
- 7) Modifiable - structure & style of requirement is such that the changes can be easily, completely & consistently made while retaining the structure.
- 8) Traceable - if the origin of requirements are clear.
Backward Traceability - depends upon each req. explicitly referencing its source in earlier documents.
Forward Traceability - depends upon each req. in the SRS having a unique ~~name~~ name or reference number.

Organisation of SRS.

- 1) Introduction
 - 1.1 scope
 - 1.2 Purpose
 - 1.3 definitions
 - 1.4 References
 - 1.5 Overview
- 2) Overall Description
 - 2.1 Product Perspective
 - 2.1.1 System Interfaces
 - 2.1.2 H/W " "
 - 2.1.3 S/W " "
 - 2.1.4 Memory constraints
 - 2.2 Product functions
 - 2.3 user characteristics
- 3) Specific Requirements
etc.

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